AQ1.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

2.1 Level 1:

1 point

Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 1 & 1 & -2 & 9 \\ -3 & 4 & 7 & 7 \end{bmatrix}$A = $\begin{bmatrix} 2 & 1 & -3 & 4 \\ 1 & -2 & 4 & 5 \\ 9 & 7 & 1 & -6 \\ 7 & -6 & 7 & 7 \end{bmatrix}$. Which of the following is true about the matrix A A?

[Hint: The (i, j)(i, j)-th entry is the entry which is at the ii-th row and jj-th column.]

- [ ] It is a $4 \times 3$4 × 3 matrix.
- [x] It is a $3 \times 4$3 × 4 matrix.
- [ ] $(2, 3)$(2, 3)-th entry of the matrix AA is 44.
- [x] $(2, 3)$(2, 3)-th entry of the matrix AA is 99.

Yes, the answer is correct.

Score: 1

Accepted Answers:

It is a $3 \times 4$3 × 4 matrix.

$(2, 3)$(2, 3)-th entry of the matrix AA is 99.

1 point

Which of the following statements is(are) TRUE?

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

- ○ Any diagonal matrix is a scalar matrix.
- ○ Scalar matrices may not be square matrices.
- ◉ Scalar matrices must be square matrices.
- ○ Any scalar matrix is an identity matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Scalar matrices must be square matrices.

1 point

Given below is the system of linear equations:

$7x_1 + 10x_2 + 12x_3 = 368x_1 + 4x_2 - 9x_3 = 114x_1 - x_2 + 3x_3 = 107$x1 + 10x2 + 12x3 = 368x1 + 4x2 − 9x3 = 114x1 − x2 + 3x3 = 10

If the matrix representation of the system of equations is $Ax = b$Ax = b, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = $\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}$, then

◉

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = $\begin{bmatrix} 7 & 8 & 4 & 10 & 4 & -11 & 2 & -9 & 3 \end{bmatrix}$, b = $\begin{bmatrix} 36 & 11 & 10 \end{bmatrix}$

$A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & 9 & 11 \\ 4 & 1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = $\begin{bmatrix} 7 & 8 & 4 & 10 & 4 & 1 & 2 & 9 & 3 & 3 & 6 & 11 & 10 \end{bmatrix}$, b = $\begin{bmatrix} 36 & 11 & 10 \end{bmatrix}$

$A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & -9 & 11 \\ 4 & -1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = $\begin{bmatrix} 7 & 8 & 4 & 10 & 4 & -11 & 2 & -9 & 3 & 3 & 6 & 11 & 10 \end{bmatrix}$, b = $\begin{bmatrix} 36 & 11 & 10 \end{bmatrix}$

$A = \begin{bmatrix} 7 & 10 & 36 & 12 \\ 8 & 4 & 11 & -9 \\ 4 & -1 & 10 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = $\begin{bmatrix} 7 & 8 & 4 & 10 & 4 & -13 & 6 & 11 & 10 & 12 & -9 & 3 \end{bmatrix}$, b = $\begin{bmatrix} 36 & 11 & 10 \end{bmatrix}$

Yes, the answer is correct.
Score: 1
Accepted Answers:

$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$A = $\begin{bmatrix} 7 & 8 & 4 & 10 & 4 & -11 & 2 & -9 & 3 \end{bmatrix}$, b = $\begin{bmatrix} 36 & 11 & 10 \end{bmatrix}$

1 point

Which of the following statements is(are) TRUE?

- [ ] Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.
- [x] Addition of two matrices is possible only if the orders of both the matrices are the same.
- [ ] Defining ABAB is possible if the number of rows in the matrix AA is same as the number of columns in the matrix BB.
- [x] Defining ABAB is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Addition of two matrices is possible only if the orders of both the matrices are the same.
Defining ABAB is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB.

1 point

Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$P = [3−17401 2−52], $Q = \begin{bmatrix} 3 & 4 & 2 \\ -1 & 0 & -5 \\ 7 & 1 & 2 \end{bmatrix}$, Q = [14−9]Q = [14−9], R = [0−310]R = [0−310], $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$

D = $\begin{bmatrix} \\ -2 \\ 4 \\ 5 \end{bmatrix}$

[Hint: If AA is a matrix of order $m \times n$m × n and BB is a matrix of order $n \times p$n × p, then the order of ABAB is $m \times p$m × p.]

- [x] The matrix PDPD is of order $3 \times 1$3 × 1.
- [ ] The matrix PDPD is of order $1 \times 3$1 × 3.
- [ ] The matrix QDQD is of order $3 \times 3$3 × 3.
- [x] The matrix QDQD is of order $1 \times 1$1 × 1.
- [x] The matrix DQDQ is of order $3 \times 3$3 × 3.
- [ ] The matrix DQDQ is of order $1 \times 1$1 × 1.
- [ ] The product QDQD is not defined.
- [x] The product QRQR is not defined.
- [x] The addition $P + Q$P + Q is not defined.
- [x] The addition $P + D$P + D is not defined.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The matrix PDPD is of order $3 \times 1$3 × 1.
The matrix QDQD is of order $1 \times 1$1 × 1.
The matrix DQDQ is of order $3 \times 3$3 × 3.
The product QRQR is not defined.
The addition $P + Q$P + Q is not defined.
The addition $P + D$P + D is not defined.

2.2 Level 2:

1 point
Choose the set of correct options.
[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$A = [a11a21a12a22] and use the given conditions to find matrix AA, for first two options.]

- [ ] If AA is a square matrix of order 2 and $A^2 = I$A2 = I, then $A = I$A = I or $A = -I$A = −I where II is the identity matrix of order 2.

- [ ] If AA is a square matrix of order 2 and A^2 = 0A2 = 0, then A = 0A = 0.
- [x] If AA and BB are square matrices of order 3 and $A + B$ = 0A + B = 0, then $B = -A$B = −A.
- [x] If AA is a scalar matrix of order 3, BB is a non-zero square matrix of order 3 and AB = 0AB = 0, then A = 0A = 0.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If AA and BB are square matrices of order 3 and $A + B$ = 0A + B = 0, then $B = -A$B = −A.
If AA is a scalar matrix of order 3, BB is a non-zero square matrix of order 3 and AB = 0AB = 0, then A = 0A = 0.

1 point

If AA is a square matrix of order 22 whose first column is denoted by C_1 C1 and second column is denoted by C_2 C2 (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$B = [b11b21b12b22] then choose the set of correct options.
[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]

- [x] The first column of ABAB is $b_{11}C_1 + b_{21}C_2$ b11C1 + b21C2.
- [x] The second column of ABAB is $b_{12}C_1 + b_{22}C_2$ b12C1 + b22C2.
- [ ] The first column of ABAB is $b_{21}C_1 + b_{11}C_2$ b21C1 + b11C2.
- [ ] The second column of ABAB is $b_{22}C_1 + b_{21}C_2$ b22C1 + b21C2.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The first column of ABAB is $b_{11}C_1 + b_{21}C_2$ b11C1 + b21C2.
The second column of ABAB is $b_{12}C_1 + b_{22}C_2$ b12C1 + b22C2.

1 point

Choose the set of correct options.
[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$A = [a11a21a12a22] and use the given condition to find matrix AA for last two options.]

- [x] There exist some real matrices AA and BB, such that $AB = BA$AB = BA.
- [ ] There do not exist any real matrices AA and BB, such that $AB = BA$AB = BA.
- [ ] There does not exist any real matrix AA, such that $A^2 = A$A2 = A.
- [x] There exists some real $2 \times 2$2 × 2 matrix AA, such that $A^2 + A + I$ = 0A2 + A + I = 0

Yes, the answer is correct.

Score: 1

Accepted Answers:

There exist some real matrices AA and BB, such that $AB = BA$AB = BA.
There exists some real $2 \times 2$2 × 2 matrix AA, such that $A^2 + A + I$ = 0A2 + A + I = 0
Suppose AA is a 3 ×× 3 scalar matrix and (1,1)-th entry of the matrix AA is 4. Suppose BB is a 3 ×× 3 square matrix such that (i, j)(i, j)-th entry is equal to $i^2 + j^2$ i2 + j2. Find the (2,2)-th entry of the

matrix $2A + B$2A + B.

16

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

1 point

Suppose $A = \begin{bmatrix} -1 & 2 \\ -4 & 7 \end{bmatrix}$A = [−1−427] and $A^2 - \alpha A + I = 0$A2 − αA + I = 0 for some $\alpha \in R$α ∈ R.

Find the value of αα.

6

Yes, the answer is correct.

Score: 1

Accepted Answers: